



Course Enrollment API

Built during OJT (On-the-Job Training)

- By: Arpit Tripathi
- Duration: 6 weeks
- Organization: Innovatech Solutions Inc.
- Tech Stack: FastAPI, SQLAlchemy, PostgreSQL
- Status: Deployed on Render



Project Introduction

The Course Enrollment API manages Students, Courses, Enrollments, and Grades. It stores data in a database and provides API endpoints for all operations, supporting CRUD tasks and grade calculation. It follows a layered structure for clarity.

A decorative background on the left side of the slide featuring a large, stylized gear with a yellow center and blue teeth, surrounded by a network of white circuit lines and smaller gears on a dark blue background.

Project Objectives

1

Student & Course Data

Store and manage student and course information efficiently.

2

Enrollment & Grades

Link students to courses and accurately add/calculate grades.

3

Data Integrity

Validate inputs and maintain clean database relationships.

4

Stable Workflows

Ensure clear outputs and reliable system operations.

High-Level Features

→ CRUD Operations

Create, Read, Update, Delete for Students, Courses, and Enrollments.

→ Grade Calculation

Automatic letter grade computation based on marks.

→ Input Validation

Ensures data quality for emails, course codes, and marks.

→ Duplicate Checks

Prevents redundant entries for emails, course codes, and enrollments.

→ Layered Architecture

Structured API design with router, service, and repository layers.

Tech Stack and Tools



FastAPI

For building high-performance backend APIs.



SQLAlchemy

ORM for database interactions and relationships.



PostgreSQL

Robust and reliable relational database.



GitHub

For version control and collaborative development.



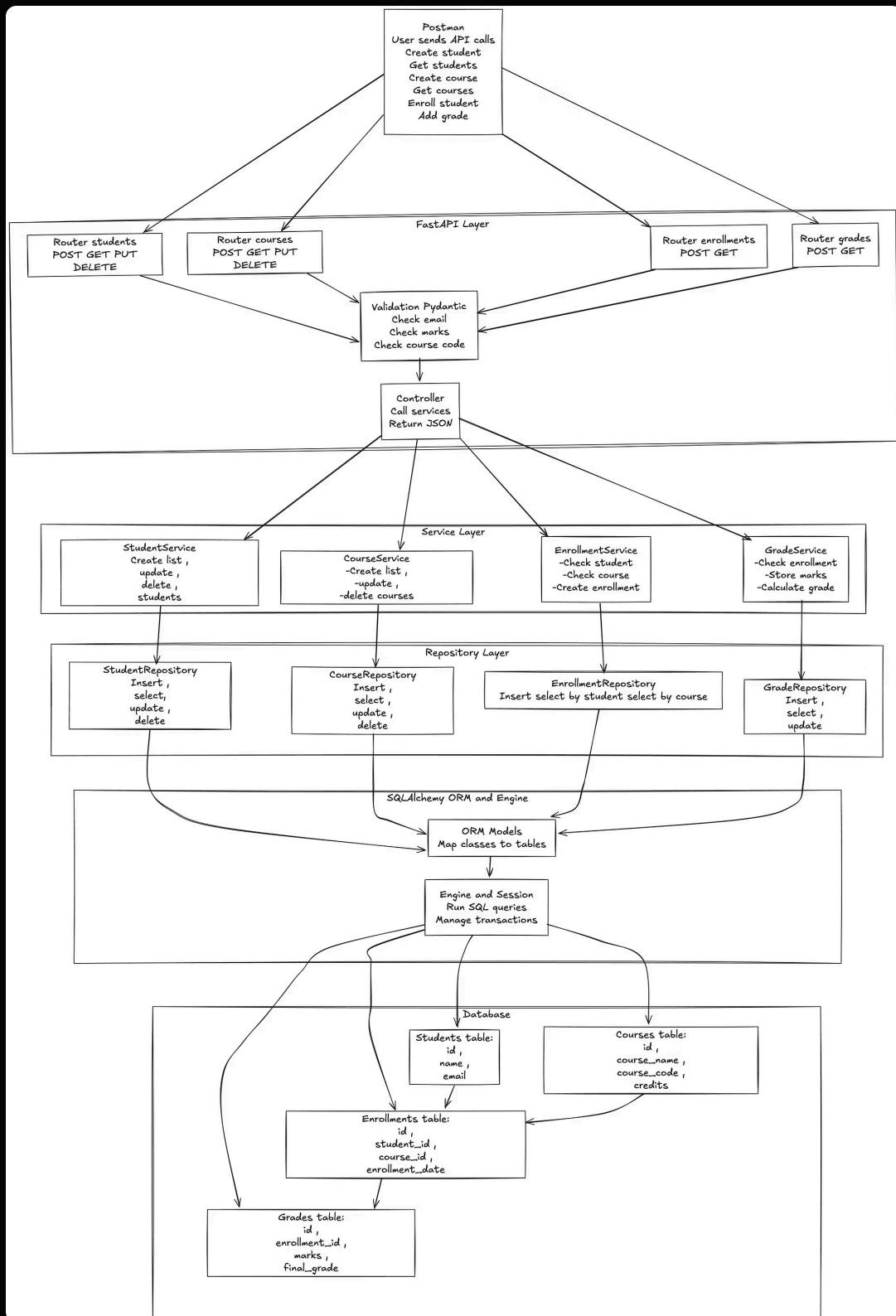
Render

Platform for seamless application deployment.



Postman

Tool for API testing and development.



HLD Overview

Database Design

Students	ID (PK), Name, Email	One-to-many with Enrollments	Unique Email
Courses	ID (PK), Code, Title	One-to-many with Enrollments	Unique Code
Enrollments	ID (PK), Student_ID (FK), Course_ID (FK)	One-to-one with Grades	Unique (Student_ID, Course_ID)
Grades	ID (PK), Enrollment_ID (FK), Marks	One-to-one with Enrollments	Marks 0-100

API Workflow

01

Create Student & Course

User initiates by creating student and course records.

02

Enrollment

Student is enrolled into a specific course.

03

Add Marks

Marks are entered for the enrolled student in the course.

04

Calculate Grade

System automatically computes the final letter grade.

05

View Data

User can view all stored and processed data.



Key Challenges & Learnings

Challenges Faced

- Learning FastAPI & SQLAlchemy
- Designing clean database schema
- Handling complex validation
- Fixing deployment errors
- Ensuring stable API flow

Key Learnings

- Building structured backends
- Using ORM relationships effectively
- Designing clean routes and services
- Effective error debugging
- Deploying live backend applications

Future Enhancements

To further mature the Course Enrollment API and enhance its functionality, several key improvements are planned.



Authentication & Roles

Implement secure user login, admin access, and role-based permissions for API operations.



Search & Filters

Enable robust searching and filtering for students and courses by various attributes like name, email, or code.



Pagination

Optimize performance for large datasets by adding pagination to student, course, and enrollment listings.



PDF Grade Reports

Generate and allow download of comprehensive PDF reports for students, detailing grades and course information.



Improved Error Handling

Standardize error messages and ensure consistent HTTP status codes for better API consumer experience.



Logging & Monitoring

Integrate logging and monitoring tools to track API activity, diagnose issues, and analyze performance metrics.



Frontend Interface

Develop a simple web-based UI to facilitate easier interaction with the API, reducing reliance on tools like Postman.



Advanced Validation

Introduce more sophisticated validation rules for marks, unique constraints, and complex relational data integrity.



Automated Tests

Build a comprehensive suite of unit and integration tests to ensure stability and reliability for future updates and changes.

Thank You!

For your time and attention to the Course Enrollment API project. I am happy to answer any questions you may have.